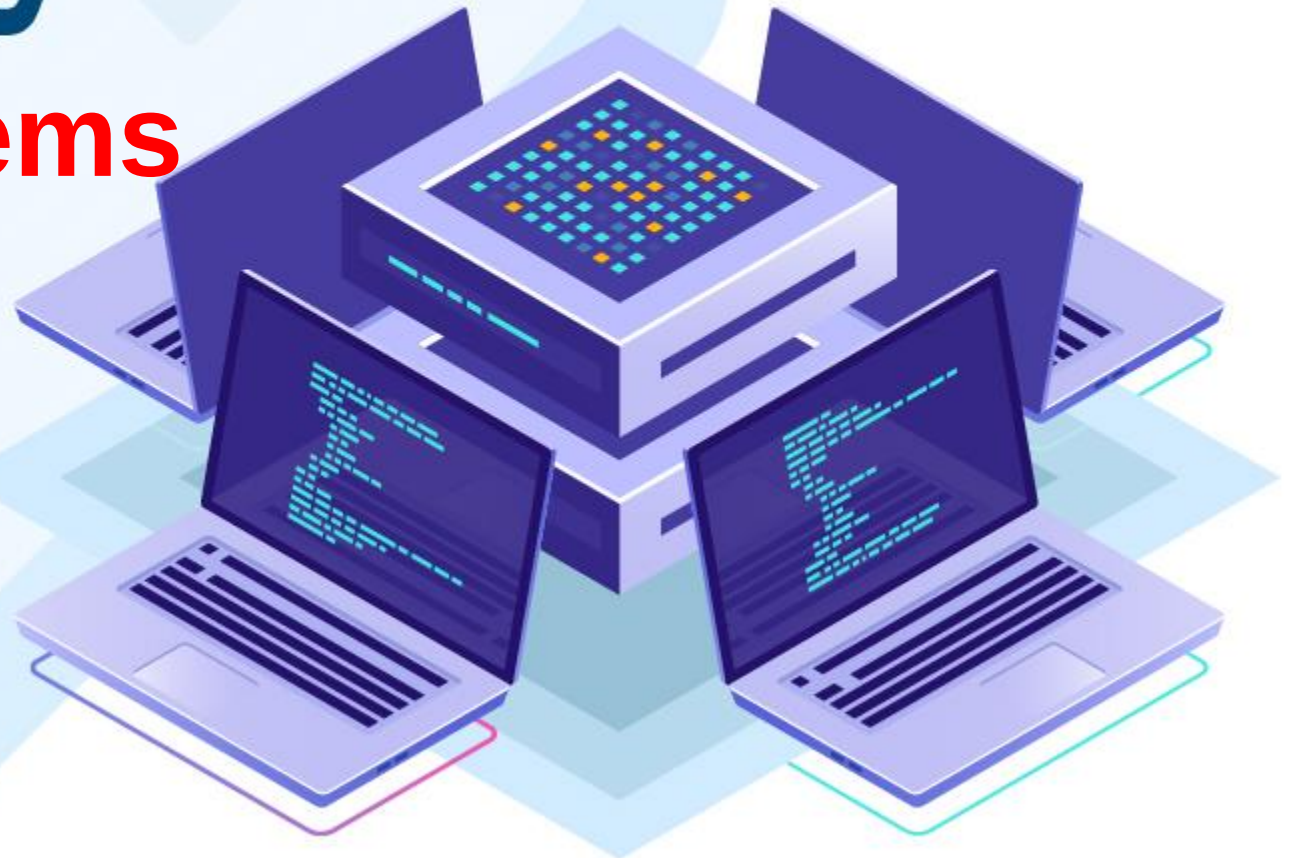


Information Technology

Fall 2023-2024
Semester 2

» Embedded Systems » LAB 1

Dr. Ahmed Ezz



الجامعة المصرية للتعليم الإلكتروني
THE EGYPTIAN E-LEARNING UNIVERSITY

Main Aim of this Lab

This Lab helps us to understand the implantation of embedded systems.



Practical Working

- We will use **ARM** controller.

Advanced **R**ISC **M**achine (**ARM**) is a group of architectures for computer processors with restricted instructions set that can be used in a variety of settings and a well-known part of a system

- We will use **Keil MDK** that is an Embedded Development tool.

Keil MDK is the complete software development environment for a range of Arm Cortex-M based microcontroller devices.

MDK includes [Keil Studio](#), the [μVision IDE](#), and [debugger](#), [Arm C/C++ compiler](#), and essential [middleware](#) components. and is easy to learn and use.

Labs Contents

1. Install the *Keil MDK*
2. Run Small experiment
3. C Language for Microcontroller
4. Build other experiments
5. Solve lecture's questions

LAB. 1

Installation of Keil & Driver

Source: Al-Azhar UNIVERSITY - Faculty of Engineering - Systems and Computers Engineering Department

Objective

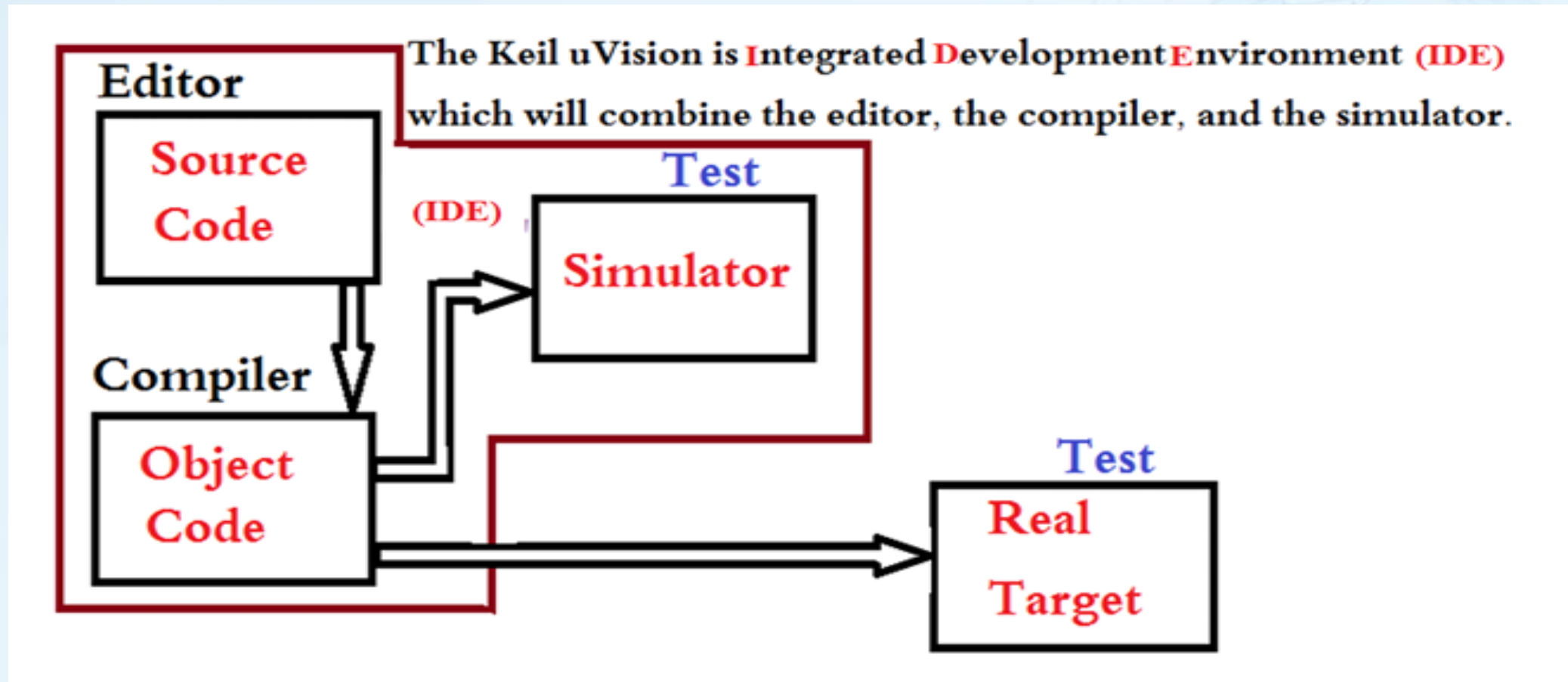
Part (a) Install the Keil μ Vision Integrated Development Environment (IDE) for the ARM. This IDE includes the editor, compiler, simulator, and debugger. The MDK474.exe (590MB) Edition does not require a serial number or license key.

• **Part (b)** Install Test EXecute and Simulate (TExaS) (60MB) . TExaS is a set of enhancements to Keil that will serve as your helper. This installation will also setup the example projects and starter files for labs.

• **Part (c)** Install the hardware device drivers for your LaunchPad (Microcontroller). We will need the LaunchPad hardware to perform this step. We can skip this step and perform the labs in simulation.

MATERIALS/EQUIPMENT NEEDED

- Keil μ Vision Integrated Development Environment (IDE) for the ARM.

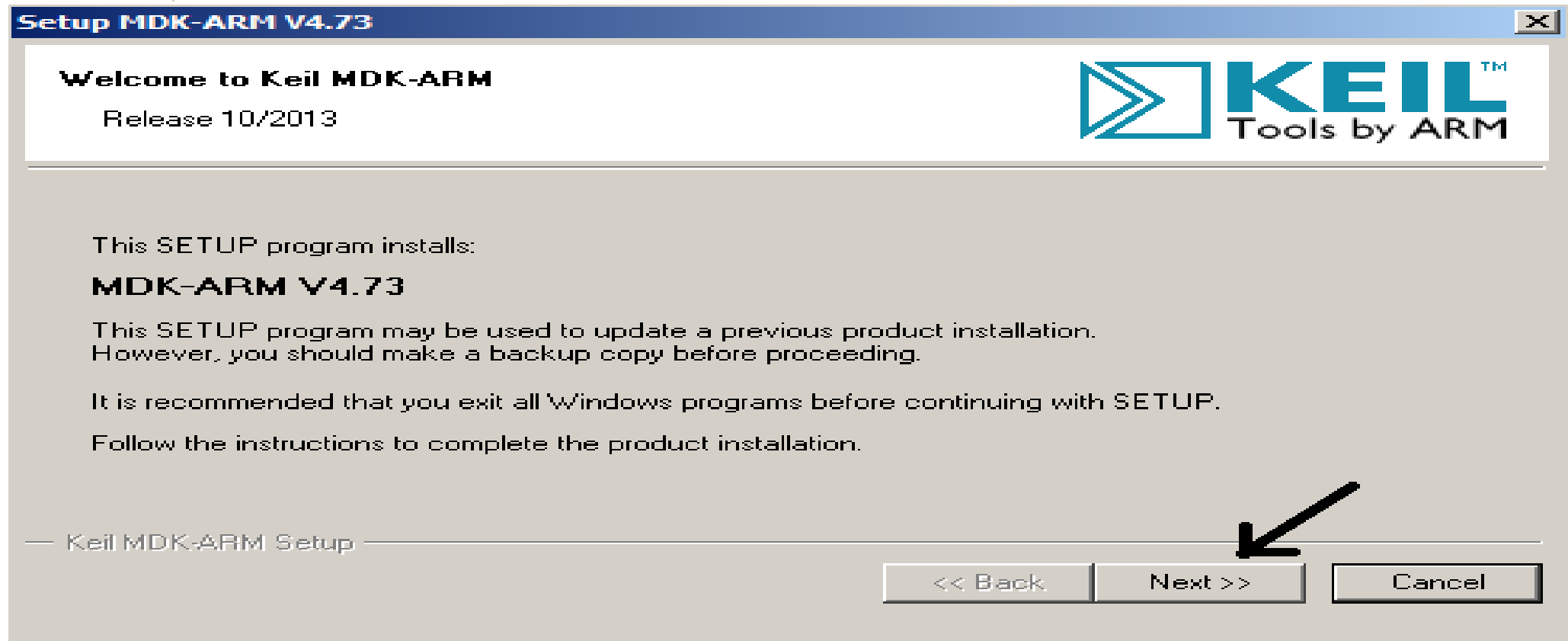


Part (a)

Install the Keil μ Vision

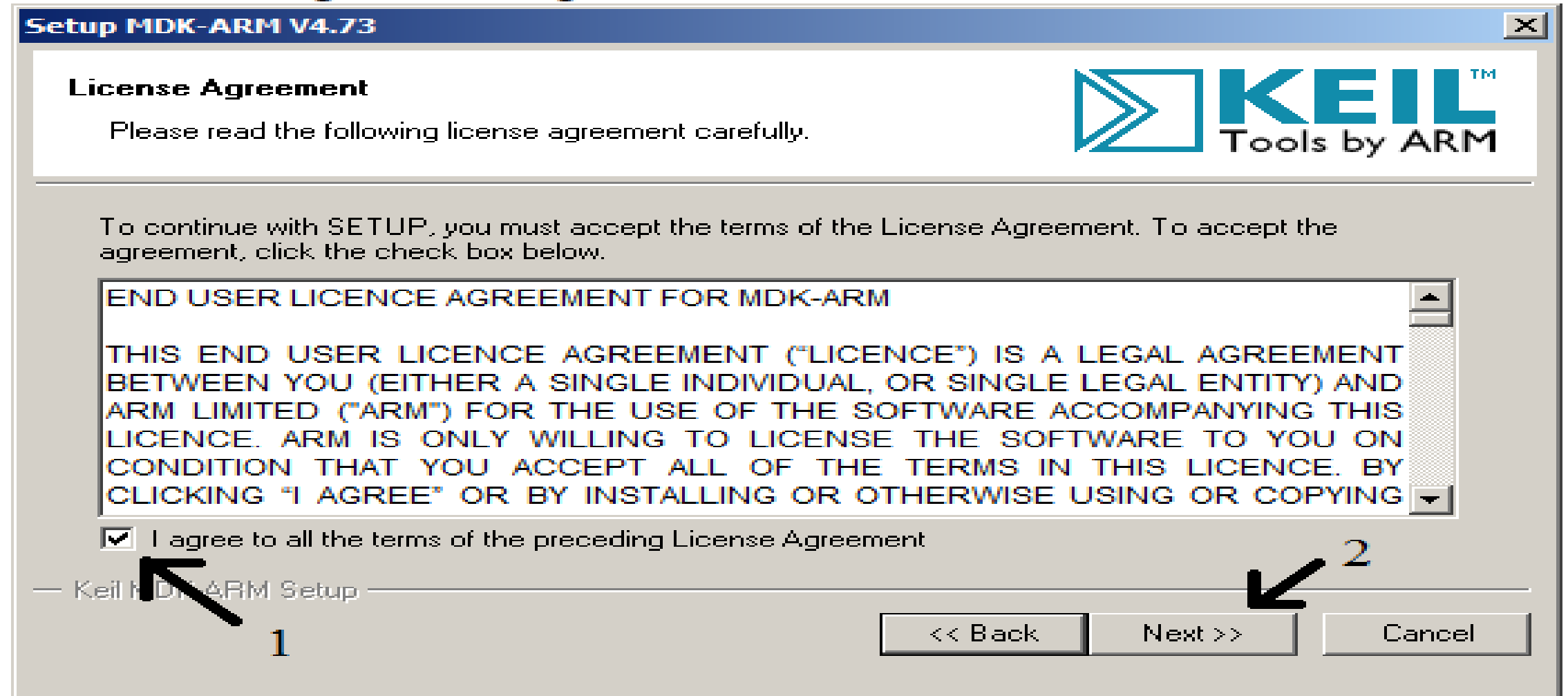
How to install Keil μ Vision for the ARM, MDK474.exe

3) Execute the **mdk474.exe** file, installing the application some place easy to find, like C:\Keil or D:\Keil. You will need to find the **Keil** directory because the TExaS example files will be loaded into this same directory. This is the first screen:



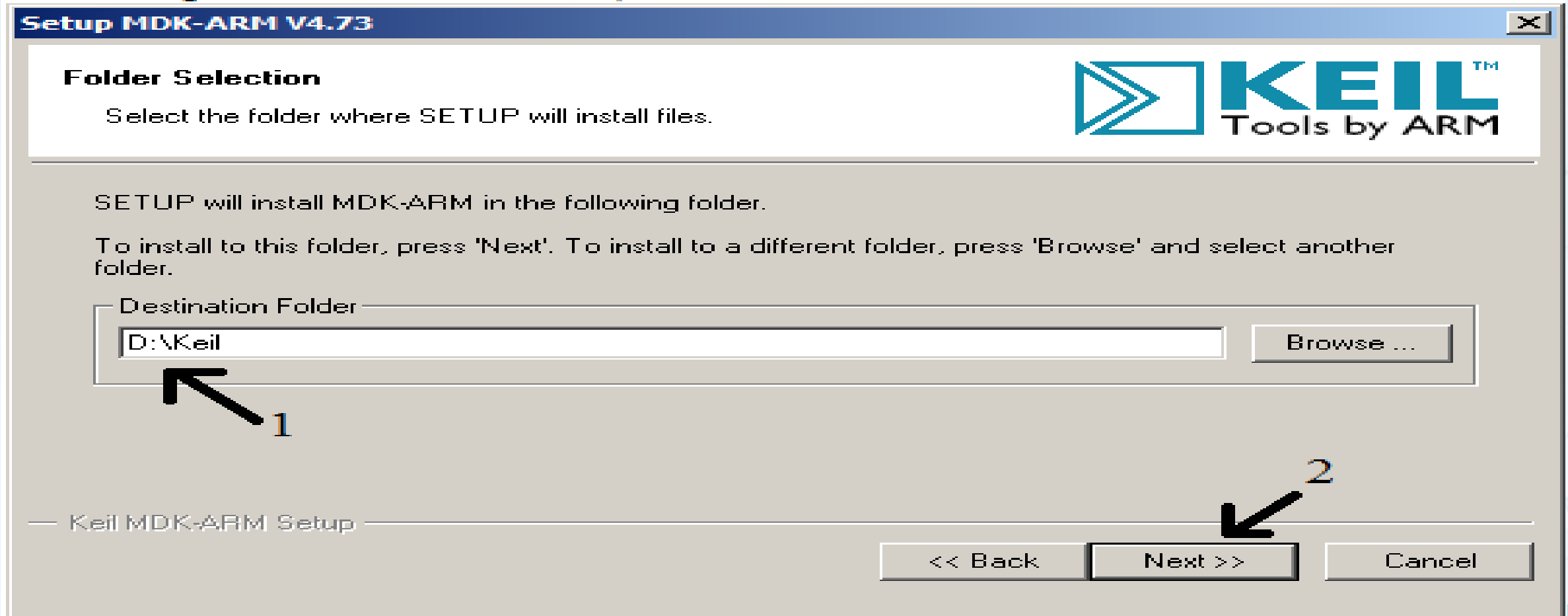
How to install Keil μ Vision for the ARM, MDK474.exe

Read license agreement, agree to terms, and click **Next**.



How to install Keil μ Vision for the ARM, MDK474.exe

Select a place to install Keil (I choose D: because there was more room on the computer on drive D) , and click **Next**.



How to install Keil μ Vision for the ARM, MDK474.exe

Update these fields with your correct information, and click **Next**.

Setup MDK-ARM V4.73

Customer Information
Please enter your information.

Please enter your name, the name of the company for whom you work and your E-mail address.

First Name:

Last Name:

Company Name:

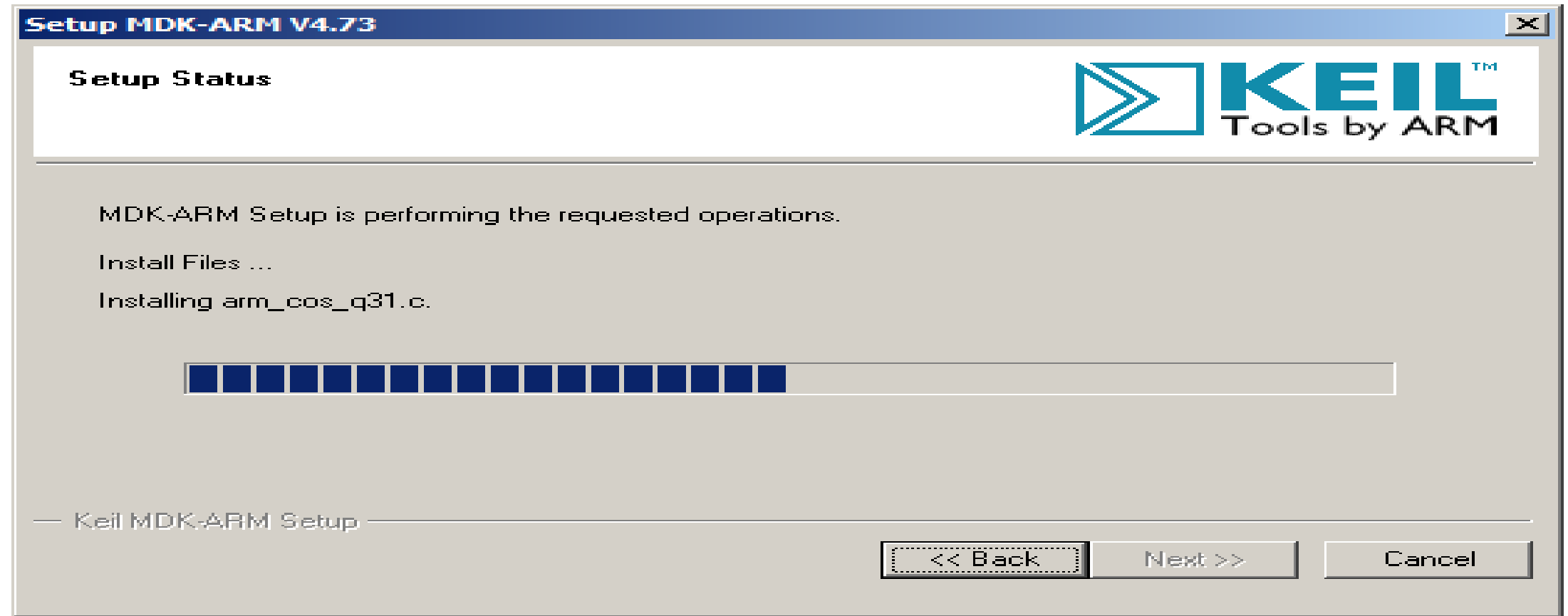
E-mail:

— Keil MDK-ARM Setup —

<< Back **Next >>** Cancel

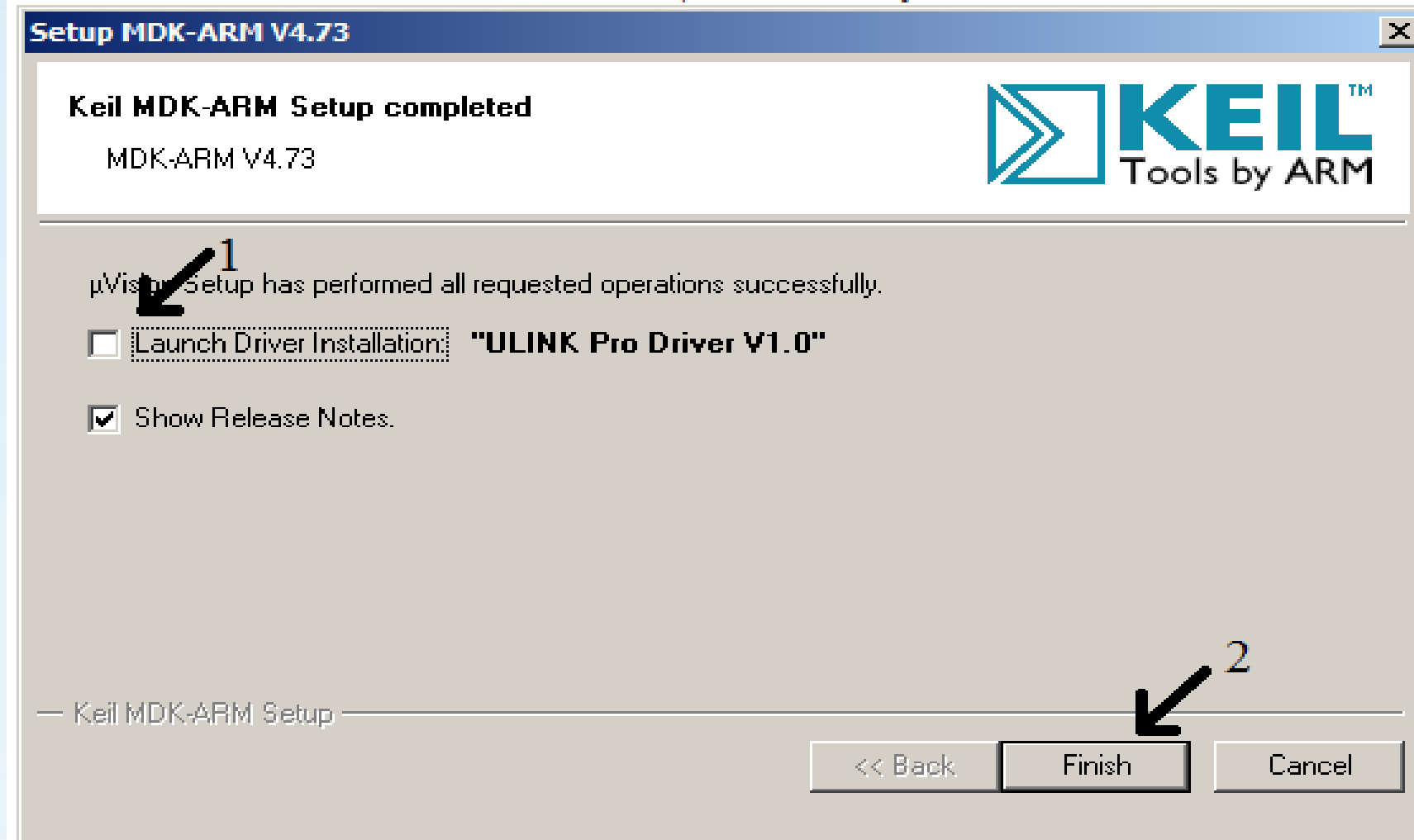
How to install Keil μ Vision for the ARM, MDK474.exe

Wait while it installs



How to install Keil μ Vision for the ARM, MDK474.exe

Deselect **ULINK Pro Driver V1.0** (the drivers you need will be installed later), and click **Finish**.

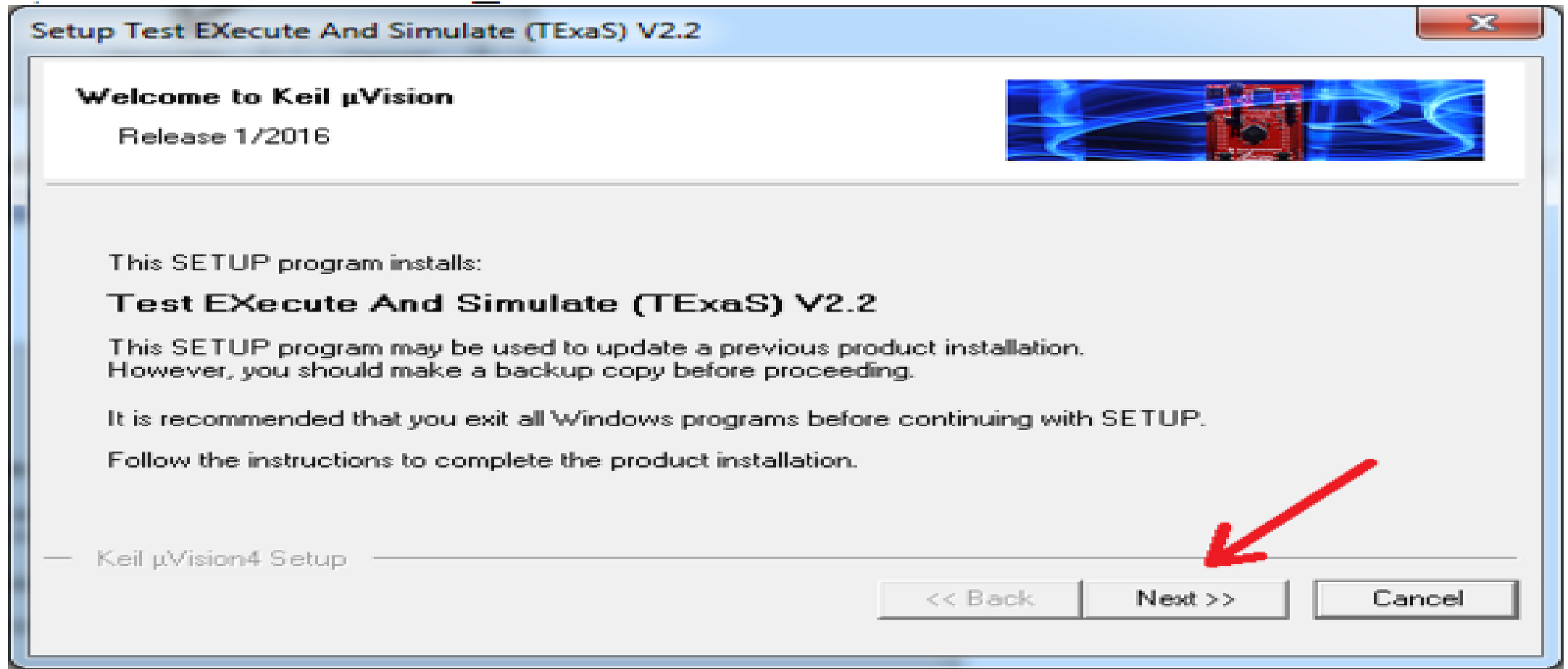


Part (b)

Install **T**est **E**xecute **a**nd **S**imulate (TExaS)

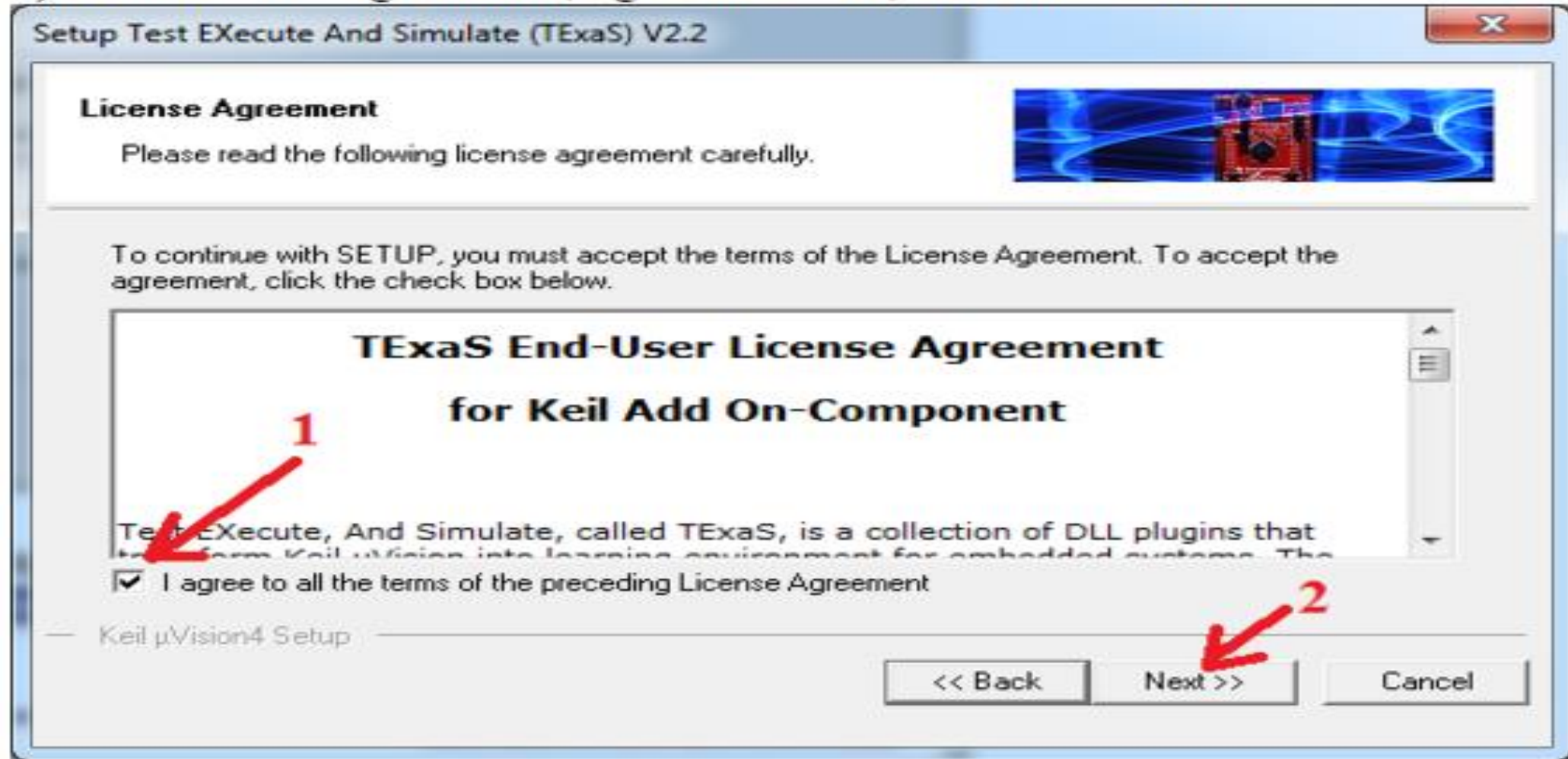
Install Test EXecute and Simulate (TExaS)

2) Execute the **TExaS Install.exe** file to install TExaS, which will add files to the Keil directory. Click **Next**.



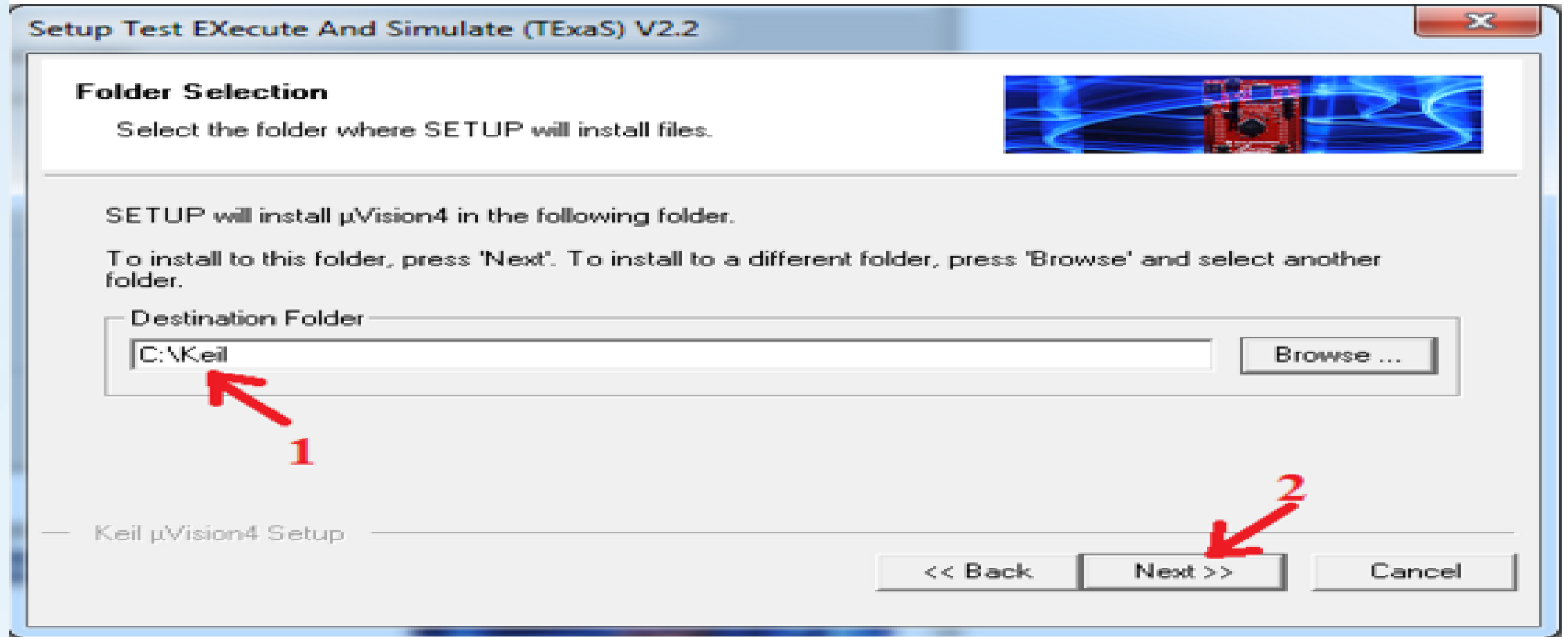
Install Test EXecute and Simulate (TExaS)

3) Read license agreement, agree to terms, and click Next.



Install Test EXecute and Simulate (TExaS)

4) Install TExaS into the same directory as Keil (this is important, you cannot install it somewhere else), and click **Next**.



Install Test EXecute and Simulate (TExaS)

5) Update these fields with your correct information, and click Next.

Setup Test EXecute And Simulate (TExaS) V2.2

Customer Information

Please enter your information.

Please enter your name, the name of the company for whom you work and your E-mail address.

First Name:

Last Name:

Company Name:

E-mail:

— Keil µVision4 Setup —

<< Back Next >> Cancel

Keil μ Vision4 Setup completed

Test EXecute And Simulate (TExaS) V2.2



μ Vision Setup has performed all requested operations successfully.

☒ Show Release Notes.

☒ Add example projects to the recently used project list.

— Keil μ Vision4 Setup —

<< Back

Finish

Cancel



Part (C)

Install the hardware device drivers.

Install the hardware device drivers.

Once Keil and TExaS are installed we will need to install the windows drivers for the Texas Instruments microcontroller board.

We must have microcontroller to do this installation

Install the hardware device drivers.

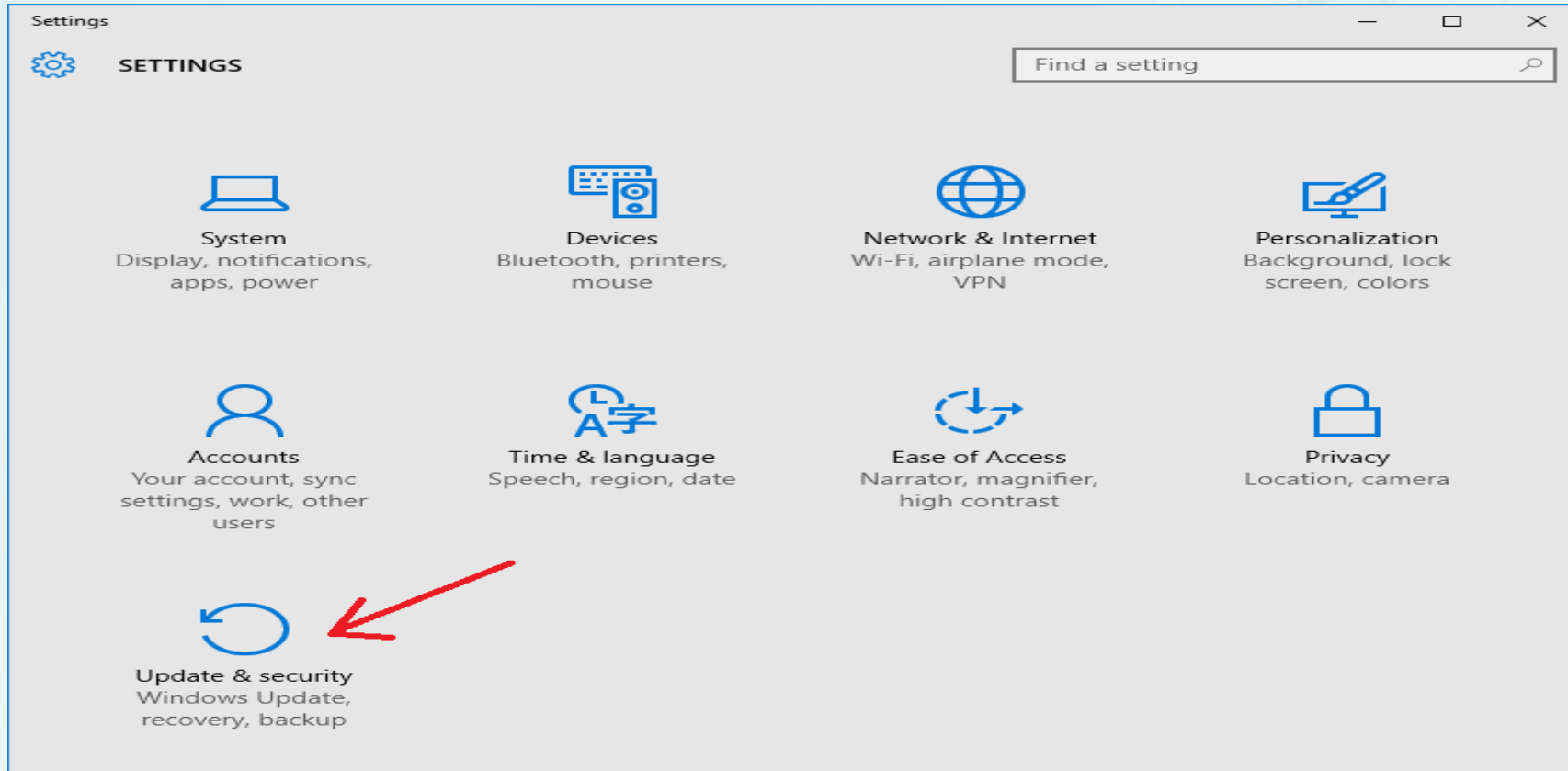
For Different windows versions

[Windows 8,](#)

[Windows 10.](#)

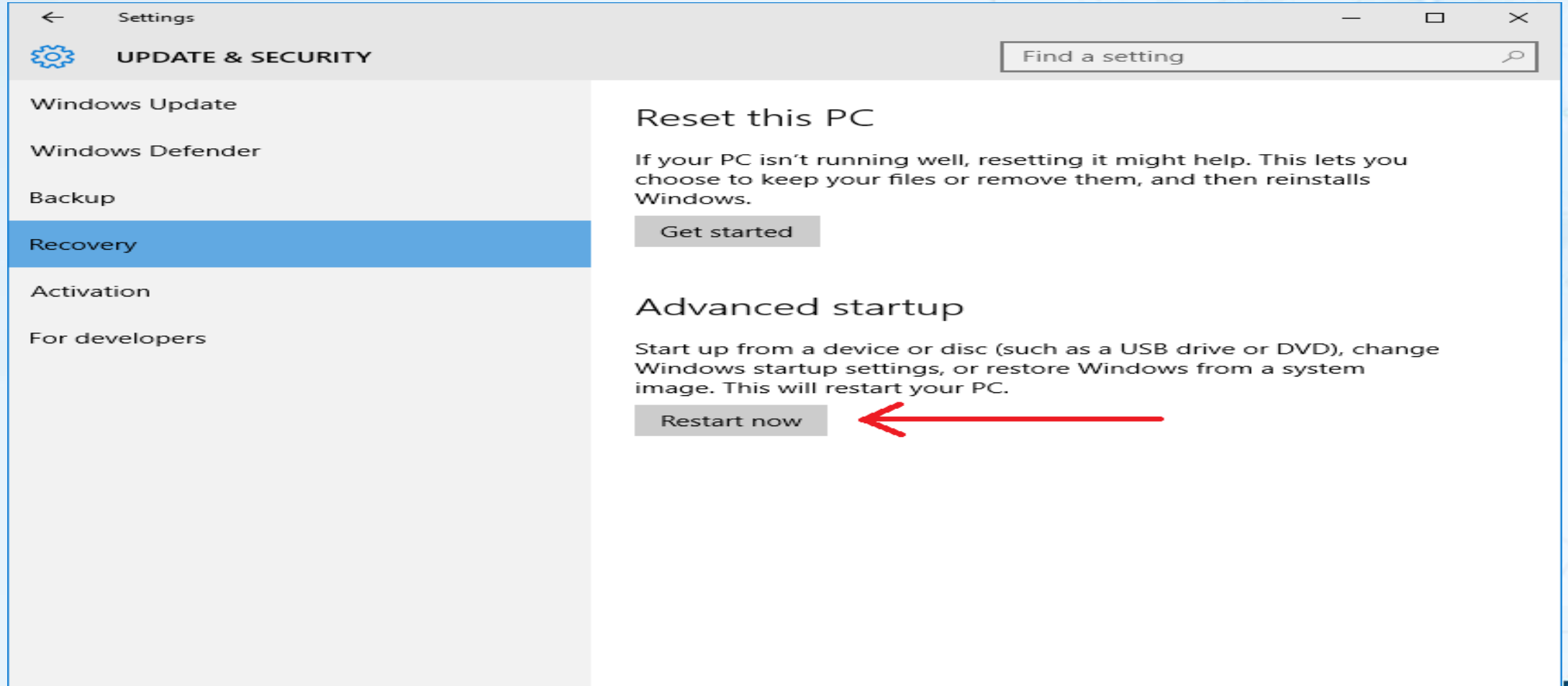
Install drivers on Windows 10.

1. Click the **Start** menu and select **Settings**. Within this window, click **Update & security**



Install drivers on Windows 10.

- 2. Click on **Recovery**, and select **Restart now** under Advanced Startup

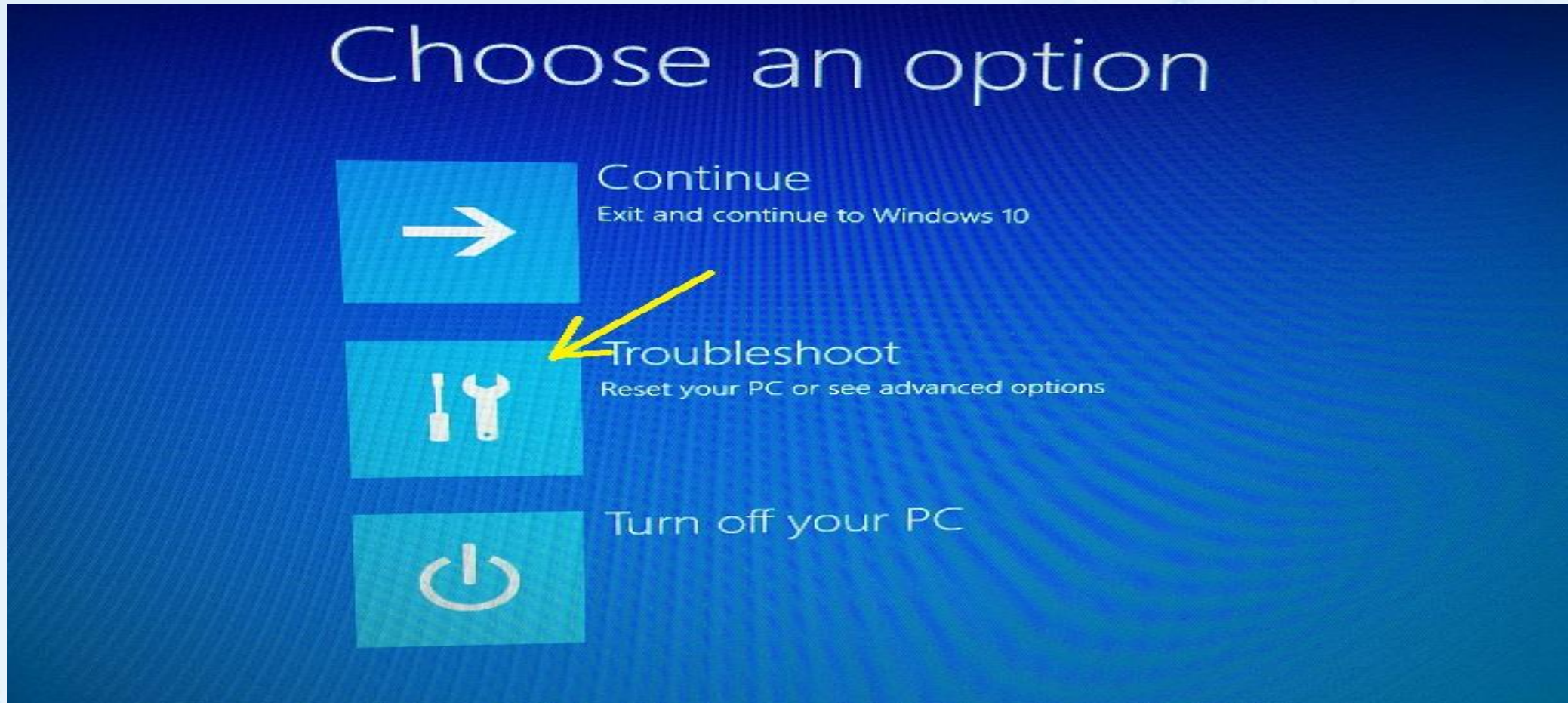


Install drivers on Windows 10.

3. The system will restart and might take some minutes to show up the boot menu. Wait for it patiently. After some time you will be prompted with a menu with following options.

...Continue ...Troubleshoot ...Turn off

Click on **Troubleshoot**



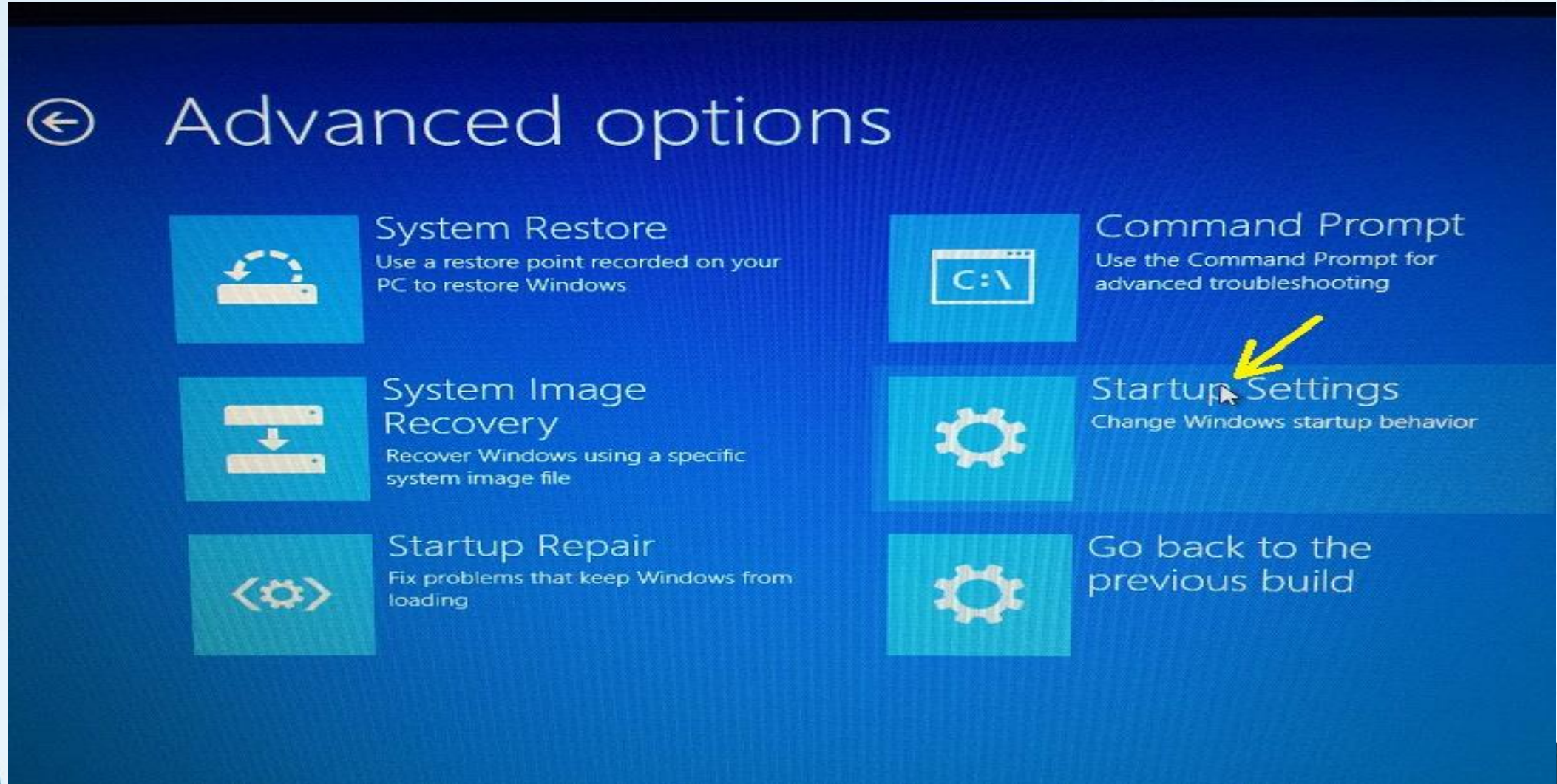
Install drivers on Windows 10.

4. Click on **Advanced options**



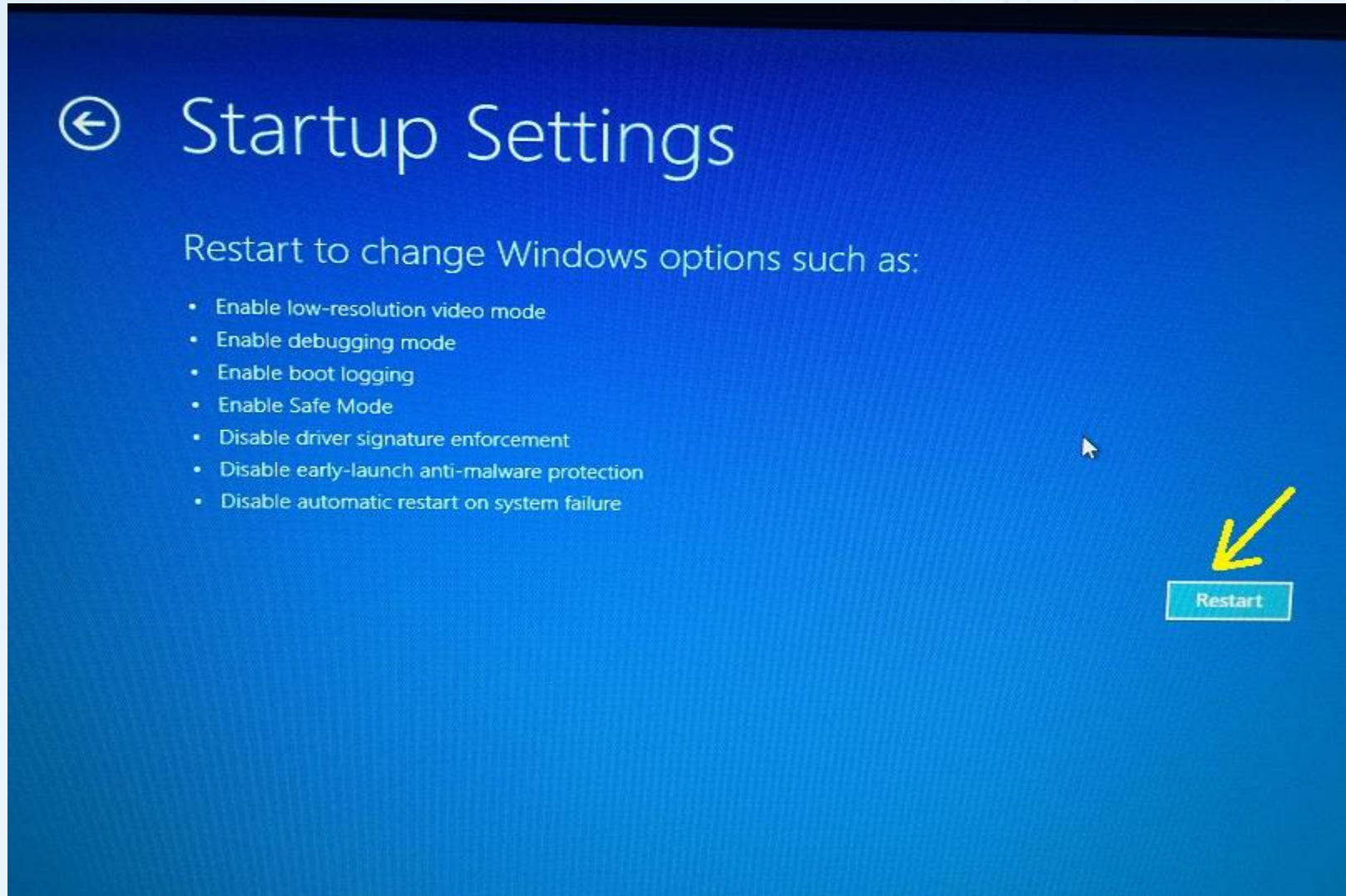
Install drivers on Windows 10.

5. Click on **Startup Settings**



Install drivers on Windows 10.

6. Click on **Restart**. Notice **Disable driver signature enforcement** will be an option.



Install drivers on Windows 10.

7. On the Startup Settings screen press 7 or F7 to **disable driver signature enforcement**. Now windows will start and you can do the installation of the driver that is not signed according to the normal procedure.

Startup Settings

Press a number to choose from the options below:

Use number keys or functions keys F1-F9.

- 1) Enable debugging
- 2) Enable boot logging
- 3) Enable low-resolution video
- 4) Enable Safe Mode
- 5) Enable Safe Mode with Networking
- 6) Enable Safe Mode with Command Prompt
- 7) Disable driver signature enforcement
- 8) Disable early launch anti-malware protection
- 9) Disable automatic restart after failure

Press F10 for more options

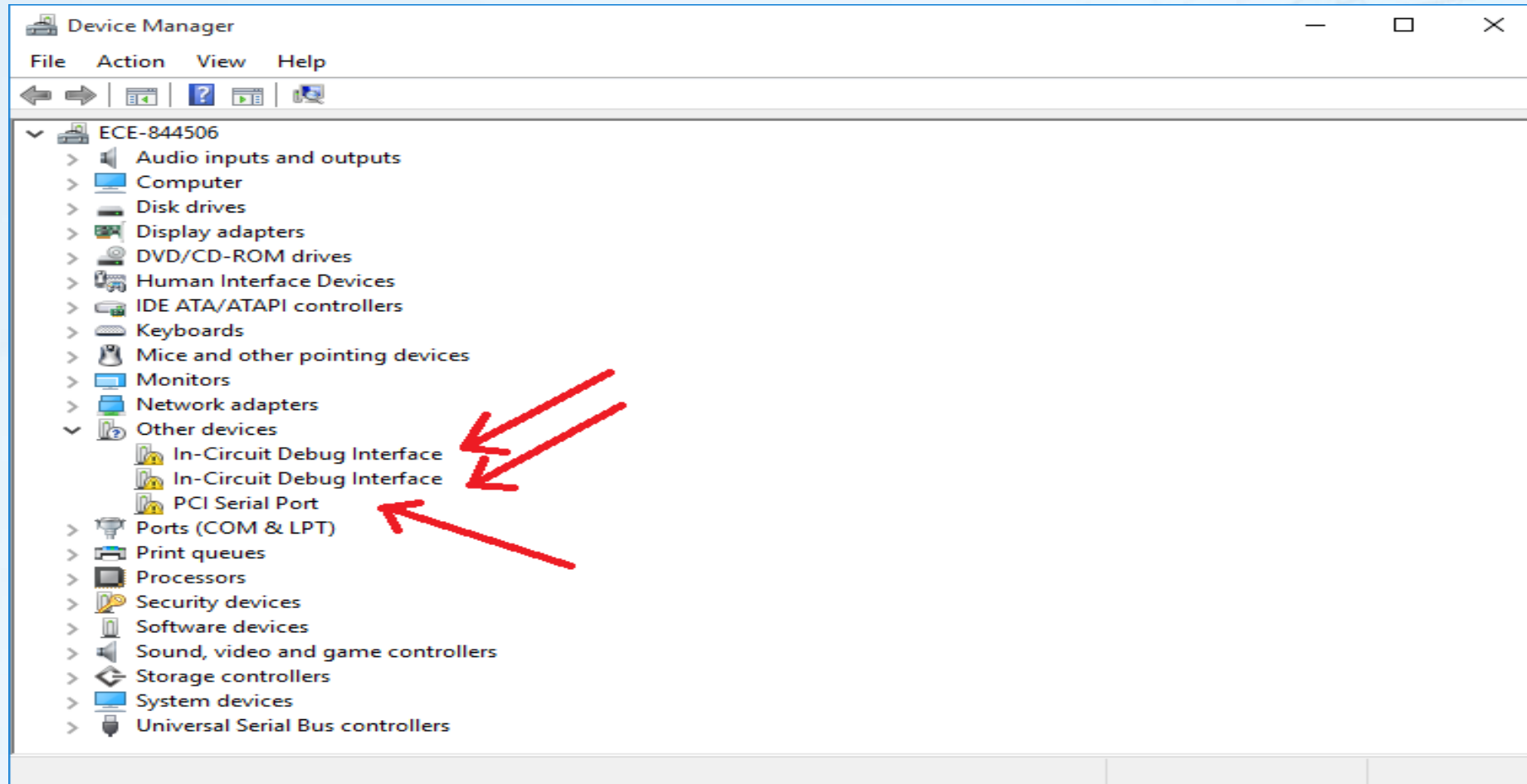
Press Enter to return to your operating system

Install drivers on Windows 10.

8. Plug in the LaunchPad into a USB port. Pick the USB port you will most likely want to use. Plug the mini-USB into the socket labelled **Debug**. Make sure the LaunchPad power is on. There is a slider switch near the USB that should be in the **Debug** position. The green LED near the USB will illuminate when power is applied.

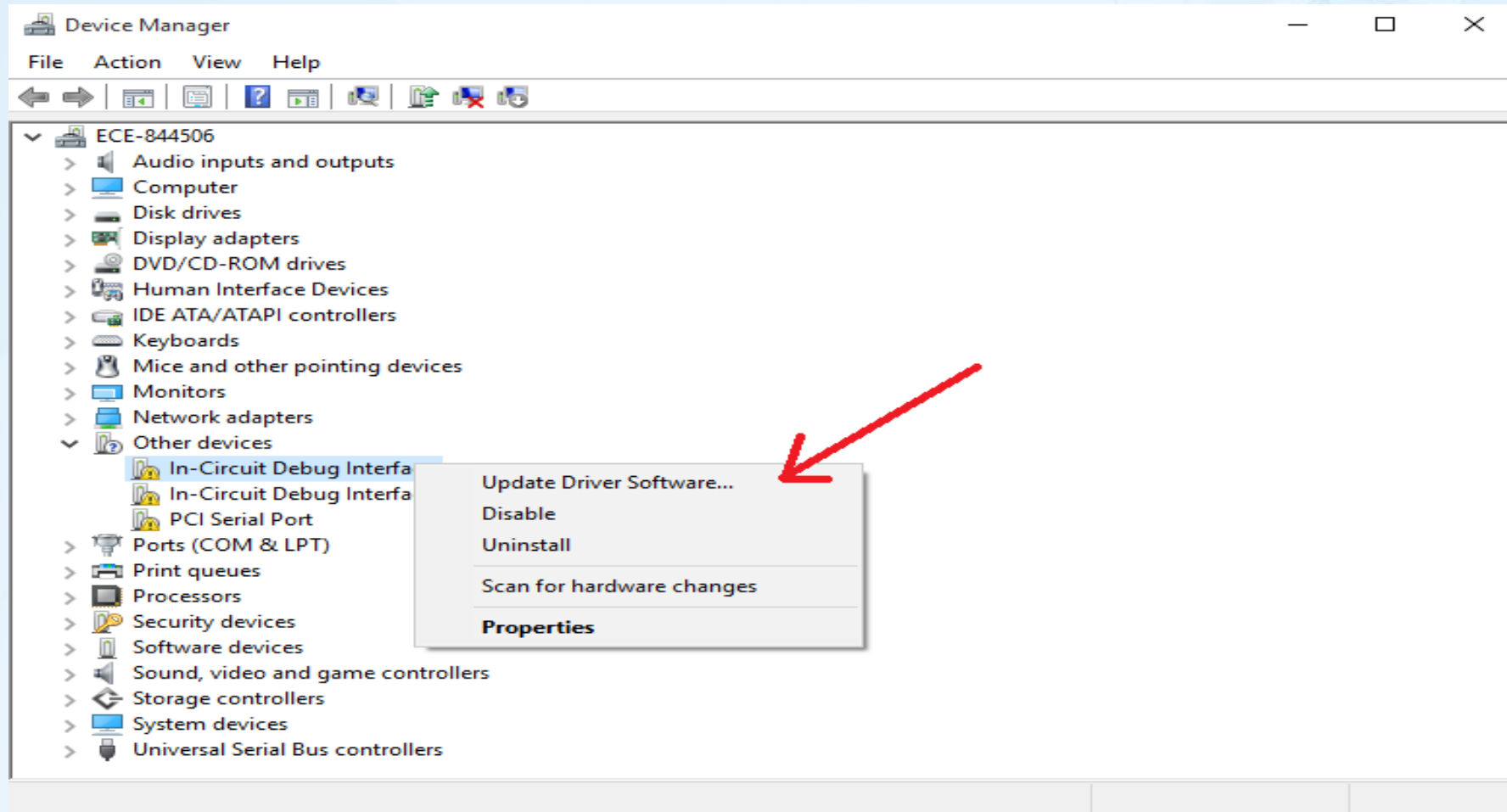
Install drivers on Windows 10.

9- Open the device manager program (click **Start**, and then in the **Search for files and programs** window type **device manager**) and you should see three In-Circuit Debug Interface devices that do not have drivers (yellow exclamation point)



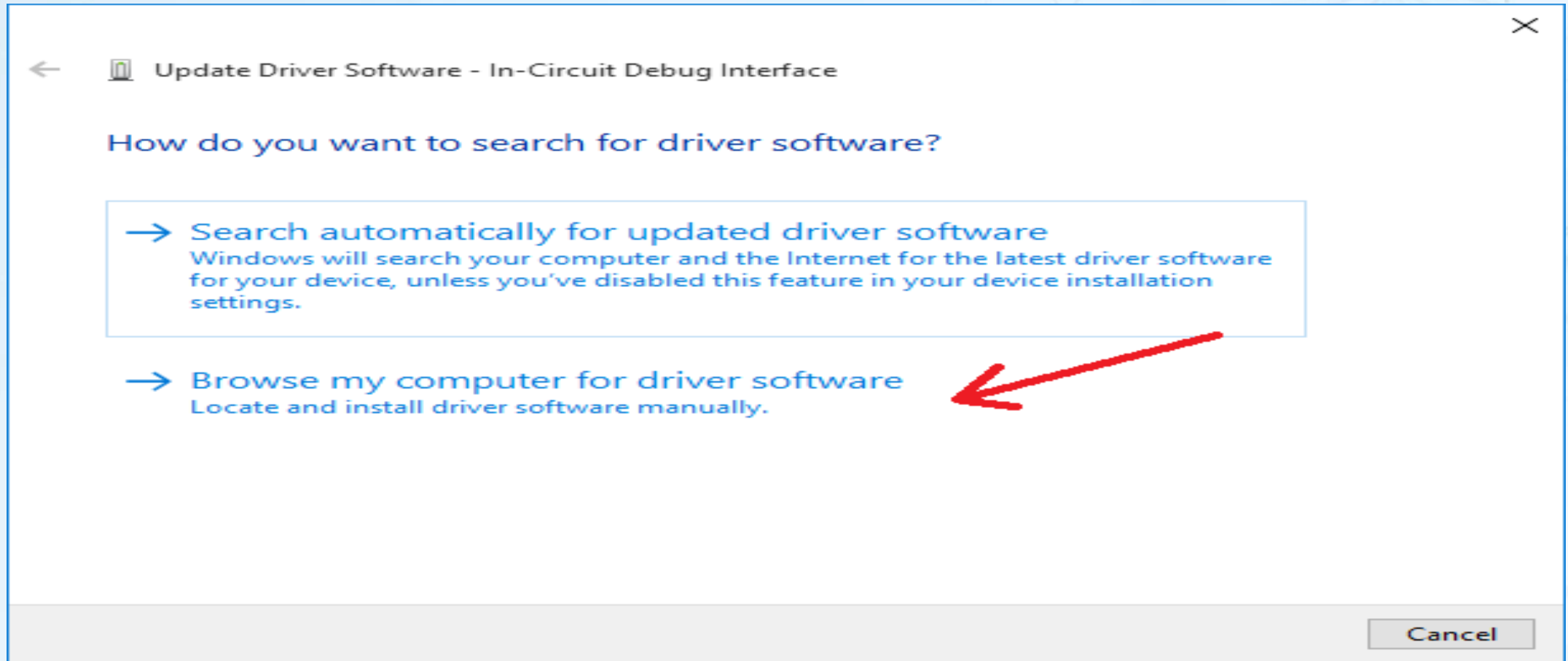
Install drivers on Windows 10.

10. For each of the three devices, first right-click on the device and then select **Update Driver Software**.



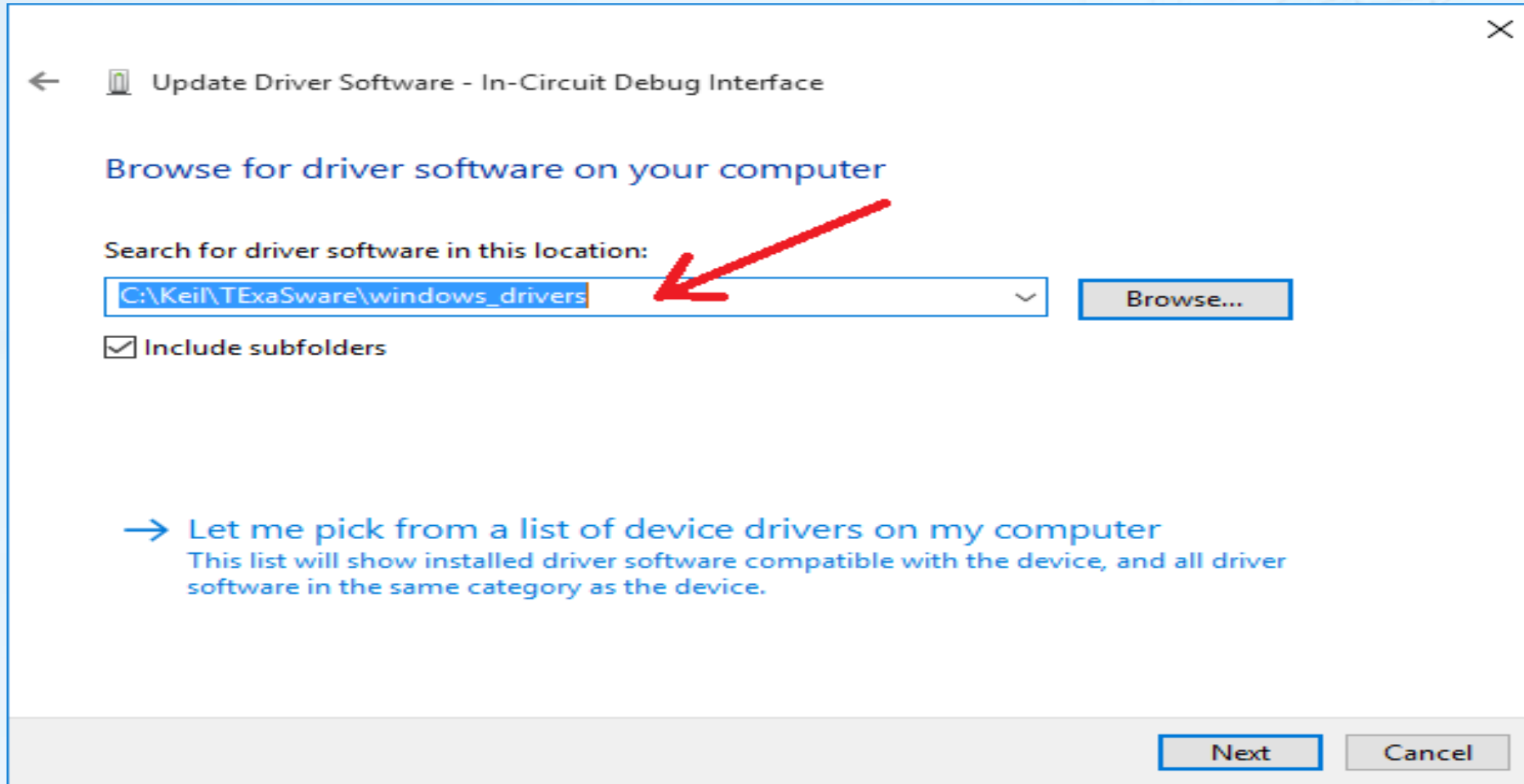
Install drivers on Windows 10.

11. You will be given a choice to *Search automatically* or to *Browse my computer*. Click **Browse my computer for driver software**.



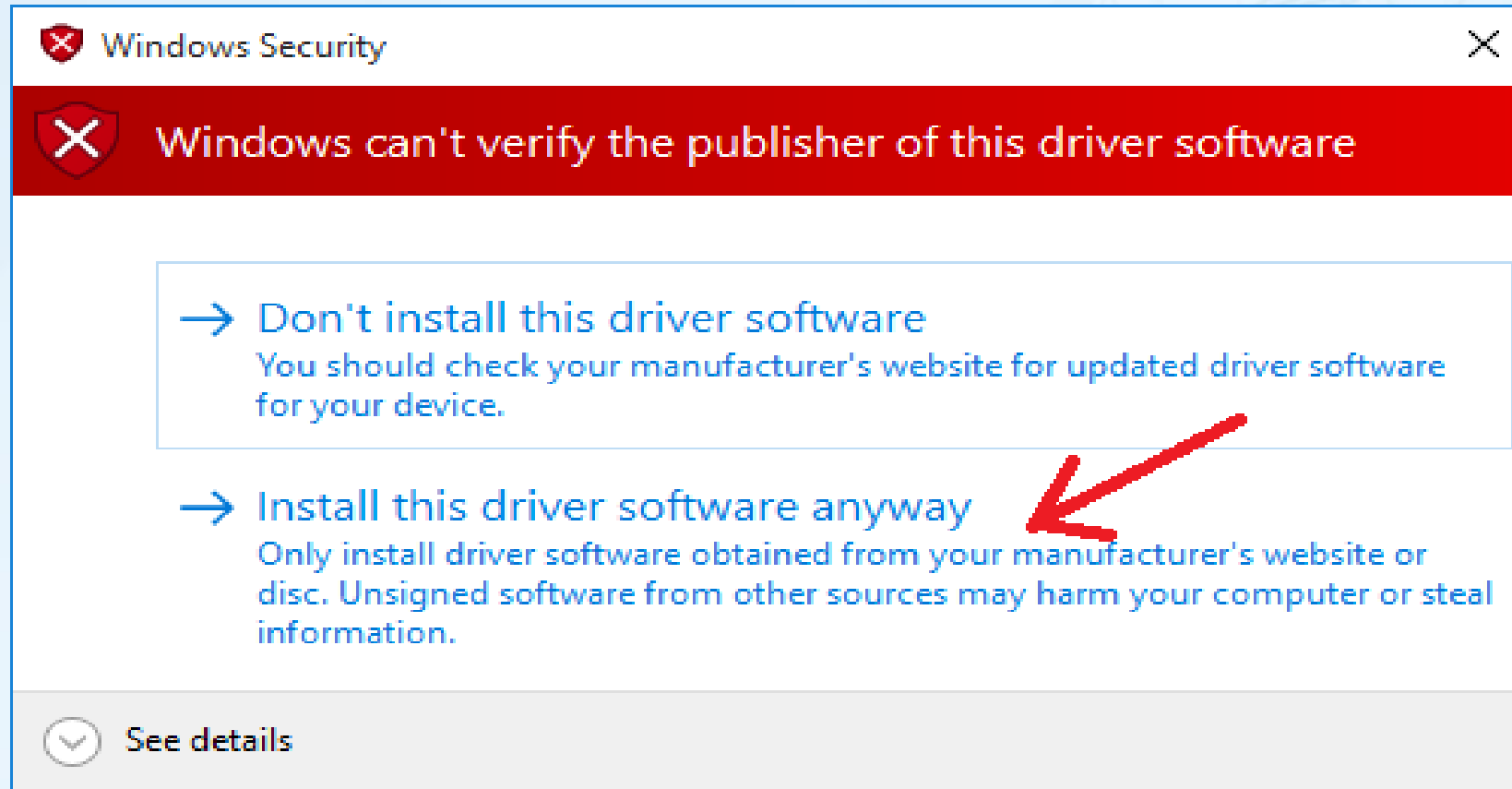
Install drivers on Windows 10.

12. Browse your computer and find the location of the **windows-drivers** folder installed with TExaS. On this computer Keil was installed on C:, but on your computer search for Keil then TExaSware and then select the **windows_drivers** folder. Click **Next**.



Install drivers on Windows 10.

13. You may get this warning. If so, click **Install this driver anyway**



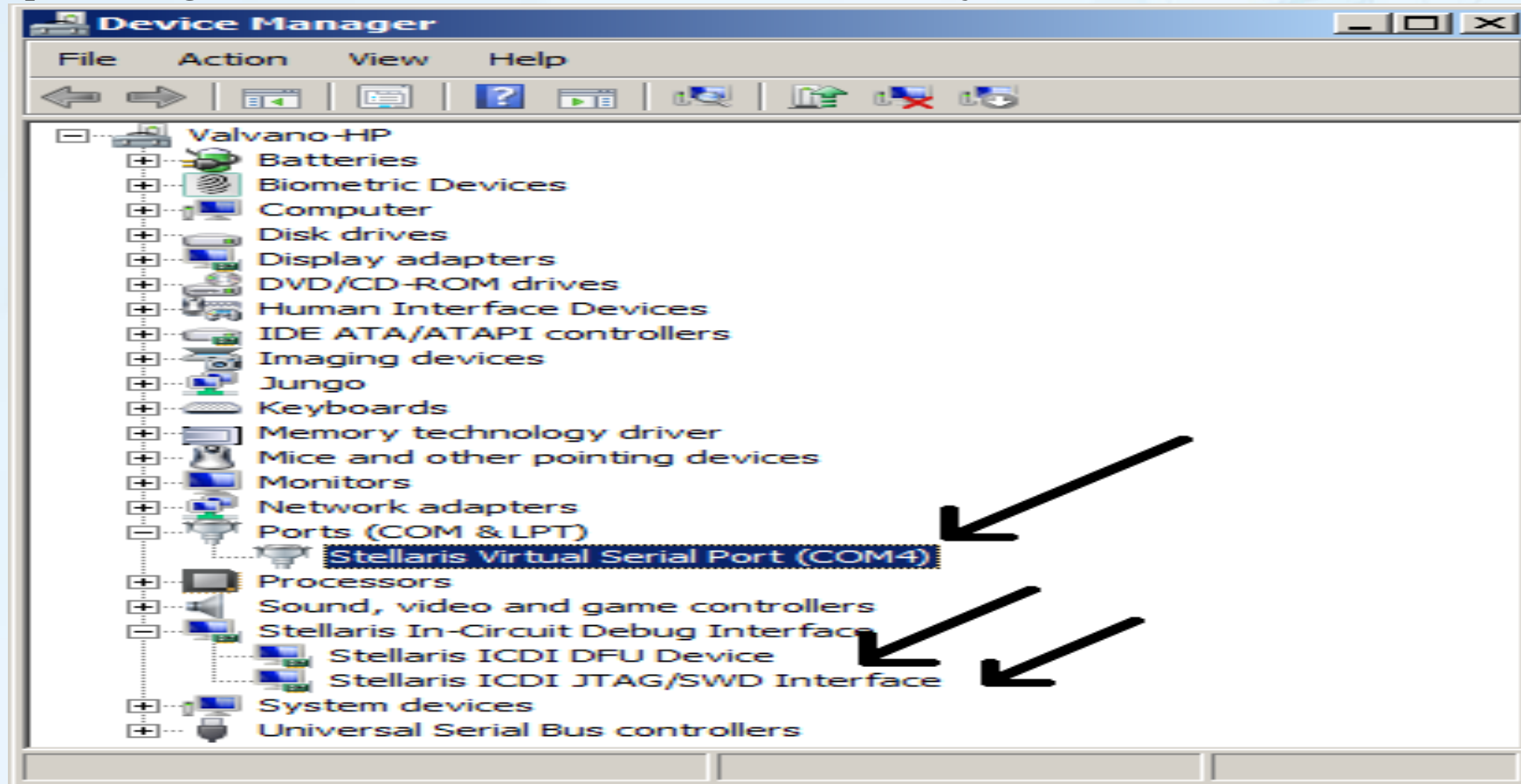
Install drivers on Windows 10.

14. You will get a success screen. Click **Next**.



Install drivers on Windows 10.

Repeat steps 10, 11, 12, 13 and 14 for the other two devices. After you have repeated the process a total of three times there should be no error flags. I.e., there should be no yellow exclamation points. However, if you do see one of the three Stellaris devices with a driver error, try right-clicking on the device with error and select **update driver**, and try the process again. Sometimes it doesn't install on the first try, but does install on the second try.



Install drivers on Windows 10.

Note that the computer shown above has one COM port (COM4 is the Stellaris board). If you see more than one COM port, write down the number associated with the Stellaris Virtual Serial Port. This number will be used in the settings command of **TExaSdisplay** you can use in some application.



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THANK YOU FOR WATCHING

QUESTIONS?

